

Da Vinci Xi Integrated Table Motion In-Service Guide

Software Version: P9 and above (da Vinci OS4 v9, v10, v11)

This document is designed to be used as a guide to facilitate the in-service on Integrated Table Motion (ITM) for the da Vinci Xi® surgical system. In-service training is recommended for every surgeon, anesthesiologist, and OR staff member who will be using this equipment.

INTUITIVE

Participant Preparation

Determine if the participants have completed the recommended online training on Intuitive Learning as indicated on the da Vinci® technology training pathway. Any activity should be demonstrated by the indicated participant. For discussions, tailor your facilitation style (Q&A vs. didactic) based on the participants' level of preparation. As the facilitator, read all notes, warnings and cautions. This training focuses specifically on the use of the Baxter® TS7000dV table and remote control in conjunction with the da Vinci Xi® system and does not encompass full training on the table as a standalone product.

In-Service Participant Roles	Participant Name
Surgeon	
OR staff	
Anesthesiologist	

Facilitator instructions

The facilitator may play any role not represented by an in-service participant.

Equipment Preparation

✓	#	System-related equipment
	1	Da Vinci Xi system with all cables connected
	4	8 mm da Vinci Xi cannulas
	1	Da Vinci Skills Model
	4	Patient cart arm drapes or sterile adapters
	1	Patient cart column drape or magnet card
	1	8 mm 0° or 30° endoscope
	3	8 mm da Vinci Xi instruments (training version if available)
	1	Da Vinci Skills Drills Upgrade Kit: 4 grommets, straps, model base

✓	#	Table-related equipment
	1	TS7000dV table
	1	TS7000dV remote control
	1	TS7000dV table connection cable
	1	TS7000dV AC power cable
	1	Low profile stirrups (if available)

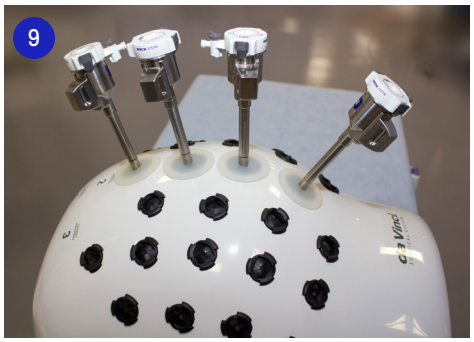
Facilitator Instructions

✓ Da Vinci Xi system and table setup

1. Ensure that all equipment from the equipment list is available at the hospital site.
2. Plug each da Vinci Xi system component into a dedicated circuit, and integrate system components via the blue cables.
3. Power on the da Vinci Xi system and pair with the table to ensure ITM has been enabled.
4. Attach sterile adapters or install arm drapes onto patient cart arms 1-4.
5. Install the column drape.
6. Place the large, silicone, grommets along midline of da Vinci Skills Model (see image 6 at right). The patient cart arms will be docked to cannulas at these port locations during this ITM in-service.
7. Orient the Skills Model such that the head-foot axis is perpendicular to the head-foot axis of the table, to accommodate the larger, silicone, grommets in the mid-line of the model (see image 7 at right).
8. Position the model in-line with the table column, and secure by attaching the models' straps to the table rails (see image 7 at right).
9. Insert cannulas through the silicone grommets, ensuring proper placement of remote centers within the body wall (see image 9 at right).
10. Unpair the TS7000dV table from the system (ensure that the connection cable for wired pairing is not attached).
11. Perform sterile stow, and power off the system.
12. Confirm that the TS7000dV table is unlocked, unplugged from AC power, and powered off.
13. Position the table with its column keypad facing away from the patient cart to facilitate access to the keypad during the in-service.







In-service topic	Activity summary	Audience		
1 Technology Overview		Surgeon	OR Staff	Anesthesiologist
Integrated Table Motion (ITM) Overview	• Define and describe basic functionality of Integrated Table Motion (ITM) feature for the da Vinci Xi system. • Describe communication between the system and table. • Specify ITM compatibility requirements. • Specify table technical specifications, including range-of-motion limits.	✓	✓	✓
Table and Remote Control Overview	• Demonstrate powering the table on/off, discuss AC power requirements, and specify best practices for use of column keypad. • Discuss key functions and components of TS7000dV remote control (LCD touch display, status bar, positioning buttons).	✓	✓	✓
Standalone Mode Overview	• Demonstrate entering Standalone mode • Discuss table use in Standalone mode		✓	✓
Demonstrate Table Adjustments in Standalone Mode	• Demonstrate locking and lowering the table • Define and discuss significance of iso-center motion, and demonstrate by adjusting Trend • Demonstrate adjustment of slide and tilt		✓	✓
2 Prepare for Intraoperative Use		Surgeon	OR Staff	Anesthesiologist
Optimizing Setup — Preparing the Table	• Define and discuss pairing • Perform wireless pairing • Lock the table, select patient orientation, and level the table	✓	✓	✓
Optimizing Setup — Preparing the Patient	• Position and secure the patient. • Place ports for procedure, discuss cannula retention considerations, and set table height at ~25-27" (65-68 cm).	✓	✓	✓
Optimizing Setup — Preparing the Patient Cart	• Deploy for docking, and drive the patient cart. • Demonstrate arm manipulation when brakes are disengaged, and review range of motion limits for patient cart arms. • Perform docking and targeting, facilitated by table height adjustment using ITM (with 1 or 2 arms docked).	✓	✓	✓

In-service topic	Activity summary	Audience		
3 Intraoperative Use		Surgeon	OR Staff	Anesthesiologist
ITM — Without Instruments	• Discuss and demonstrate pre-ITM checks. • Demonstrate turning ITM on and off (without installed instruments). • Review safety considerations before, during, and after ITM.	✓	✓	✓
ITM — With Instruments	• Demonstrate attempt to turn on ITM with 3 instruments installed. • Demonstrate turning on ITM with 2 instruments installed.	✓	✓	✓
Intraoperative Adjustments	• Describe and demonstrate conditions requiring adjustment of port position.	✓	✓	✓
Pausing and Canceling ITM	• Discuss and demonstrate how to pause and cancel ITM.	✓	✓	✓
4 Troubleshooting		Surgeon	OR Staff	Anesthesiologist
Troubleshooting	• Demonstrate resolving situation in which pairing is lost • Demonstrate pairing via a wired connection (optional) • Discuss gaining immediate patient access • Additional troubleshooting resources	✓	✓	✓

1 Technology Overview

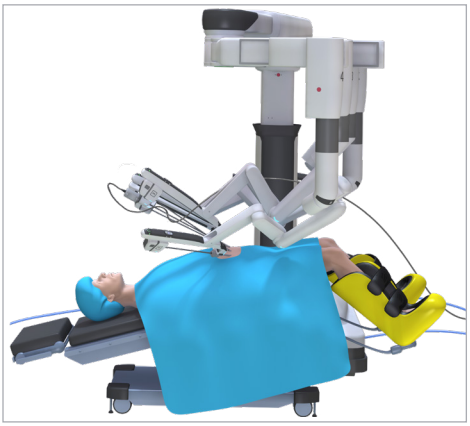
Integrated Table Motion (ITM) Overview

Learning objectives

- Define the Integrated Table Motion (ITM) feature, its benefits, and situations in which ITM is unavailable
- Describe how the da Vinci Xi system communicates with the TS7000dV table
- Specify ITM compatibility requirements
- Define key technical specifications for the TS7000dV table, including range-of-motion limits

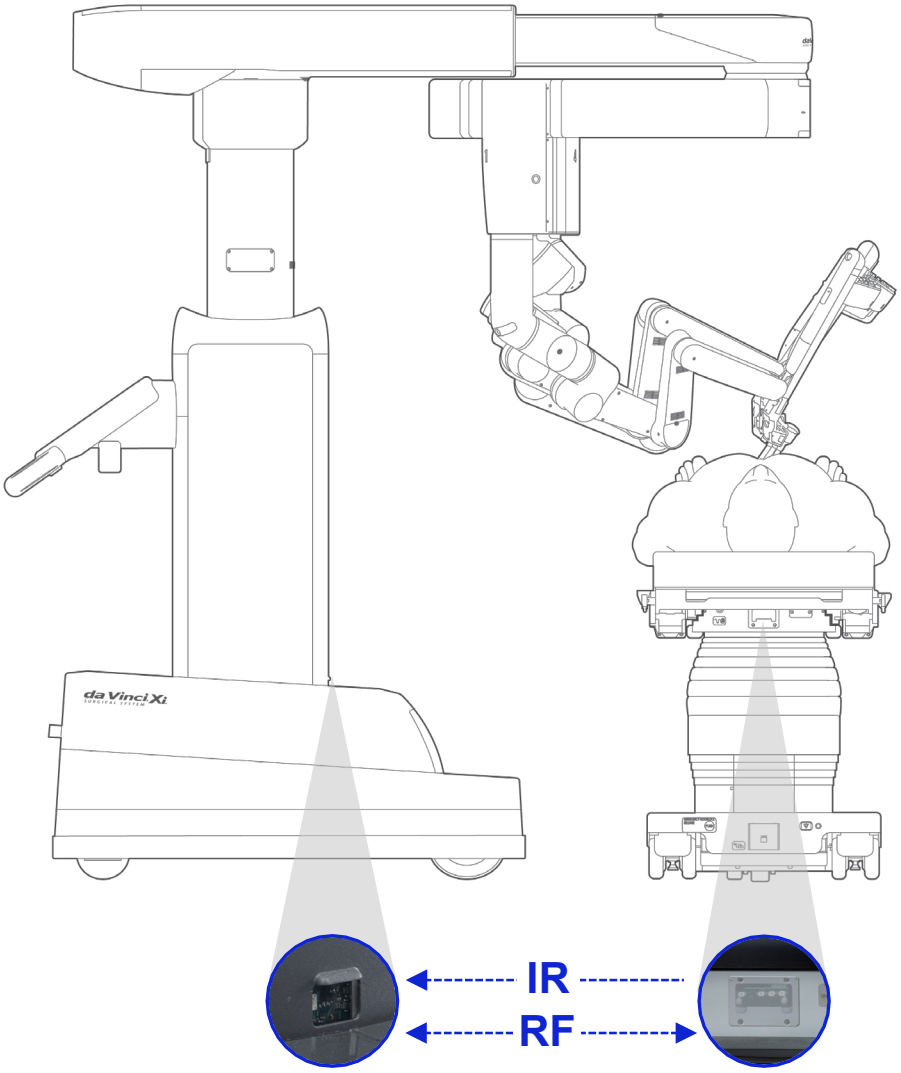
ITM Definition Discussion

- Integrated Table Motion (ITM) enables the OR team to dynamically position the patient without undocking the da Vinci Xi patient cart arms from the patient during surgery.
- During surgical use, the da Vinci Xi's patient cart arm joints (i.e. flex joints, slide joints) are locked to allow precisely controlled instrument motion.
  - When ITM is turned on, the brakes on the arm joints release, unlocking the arms and allowing them to follow the patient's movement as the table is repositioned.
  - When ITM is turned off, the brakes on the arm joints re-engage, locking the arms back in place and allowing normal surgical use to continue.



Communication Between System and Table Discussion

- Infrared (IR) and radio frequency (RF) signals travel between sensors at the top of the table column and a sensor at the bottom of the patient cart center column.
  - Infrared (IR) signals allow for table detection and pairing.
  - Radio frequency (RF) communication link allows for coordinated movements commanded by the remote control.
- To avoid disconnection during a procedure, care should be taken not to block any of the infrared (IR) sensors.
  - Drapes do not block the IR signal.



Technology Overview

Compatible System Requirements Discussion

System requirements:

- **Compatible system: da Vinci Xi surgical system**
- ITM feature enabled by an Intuitive field service engineer
  - If the ITM feature is enabled, the table tab on the vision cart touchscreen displays the pairing status as either **PAIRED** or **NOT PAIRED**
  - If the ITM feature is not enabled, the Table tab will display the pairing status as **PAIRING NOT POSSIBLE**

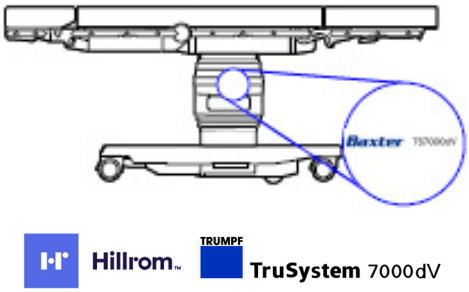


Warning: Not all systems and tables are compatible with Integrated Table Motion. Undock from the patient before performing table movement with incompatible systems and operating tables to prevent patient harm.

Technology Overview

Compatible Table and Remote Requirements Discussion

Table and remote control requirements:



- **Compatible table: Baxter TS7000dV table**
  - Verify compatibility by identifying the TS7000dV label on the table column
  - Previous branding:
    - Trumpf Medical TruSystem 7000dV Surgical Table
    - Hillrom TS7000dV Surgical Table

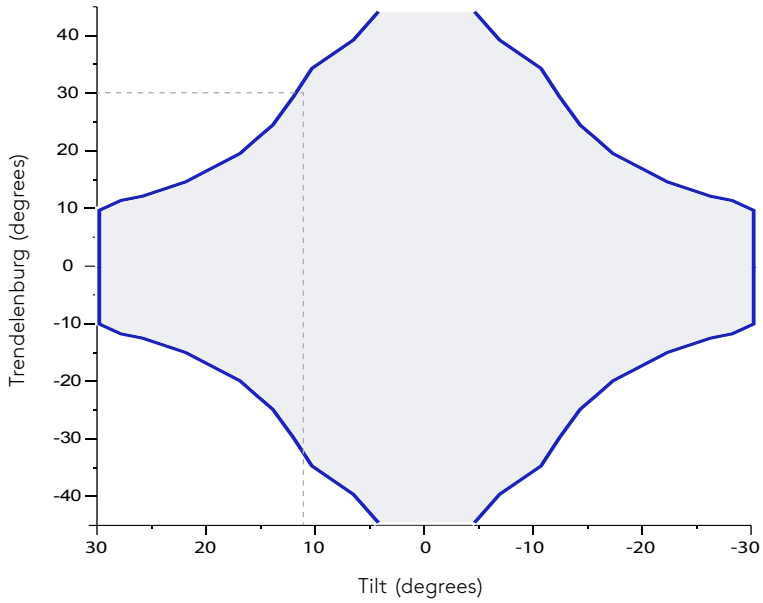


- **Compatible remote control: Baxter TS7000dV remote control**
  - Verify compatibility by identifying the DA VINCI (dV) button on the remote control
  - Previous branding:
    - Trumpf Medical TruSystem 7000dV Remote Control
    - Hillrom TS7000dV Remote Control

Table Technical Specifications Discussion

- Maximum Trendelenburg: 45° (Trend or reverse Trend)
- Maximum tilt: 30° (right or left)
- Height range: 22.8-45" (58-114.5 cm)
- Table weight limit: 1000 lbs (450 kg)

The maximum Trend and tilt angles of 45° and 30°, respectively, represent independent extremes of motion. Not all combinations of Trend and tilt are achievable simultaneously. For instance, with a Trend angle of 30°, the maximum possible tilt angle is approximately 12° (see graph at right).



Technology Overview

Table and Remote Control Overview

Learning objectives

- Discuss and demonstrate ways to power on/off the TS7000dV table
- Describe best practices and safety considerations for column keypad adjustments and table battery charging
- Discuss functionality of the TS7000dV remote control

Facilitator notes

- Ensure that the da Vinci Xi system is powered off
- Ensure that the TS7000dV table has a charged battery and is powered off and unplugged from AC power
- Note that the activity's flow (i.e. turning the table on, off, and on again) is designed to facilitate demonstration

Power the Table ON/OFF Activity

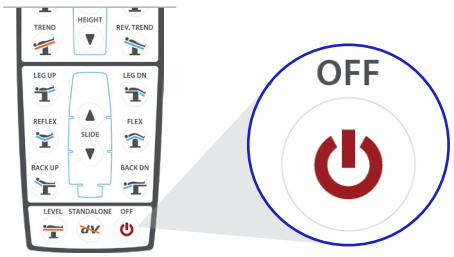
OR staff and anesthesiologist

1. Manually power on the TS7000dV table using the **ON** button on its column keypad.
2. Power off the table by pressing the **OFF** button on the remote control.
3. Automatically power on the table by connecting the table to AC power.
   a. Ensure that the table is connected to a dedicated circuit (separate from those used for the da Vinci Xi patient cart, vision cart, surgeon console, and ERBE).



Column keypad

Note: Once a cannula is docked to the da Vinci Xi surgical system, users should only adjust the table using the TS7000dV remote control, and never the column keypad, as the column keypad bypasses communication between the table and the system.



Remote control

Warning: Undock from the patient prior to using the table column controls to move the table. When docked, unintended movement of the patient relative to the cannula remote center could cause port-site damage or other injury.

Power the Table ON/OFF Discussion

- Always charge the table battery overnight.
- Confirm that the table is close enough to an outlet in the event that it needs to be plugged in mid-procedure. This allows for use of AC power without having to undock the patient cart and move the table.



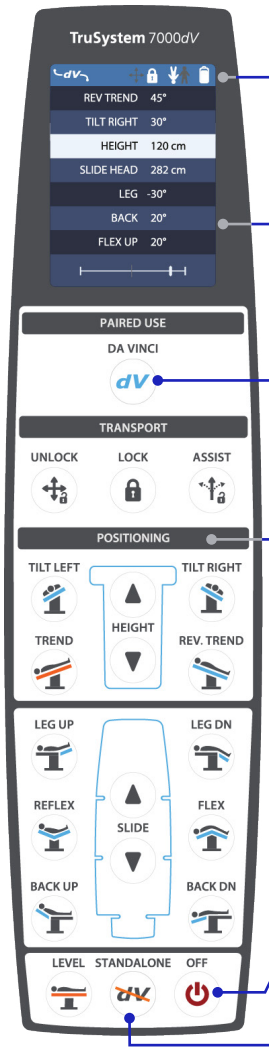
Battery indicator

Technology Overview

Key Functions of the TS7000dV Remote Control Discussion

Non-sterile OR staff and anesthesiologist

- The TS7000dV remote control facilitates setup and intraoperative use by:
  - Indicating available functions by illuminating respective positioning buttons
  - Providing feedback messages to the user
- Components of TS7000dV remote control:
  - Status bar
  - LCD touch display
  - Dismiss status messages by touching the display's lower right-hand corner
  - **DA VINCI (dV)** button
  - Positioning buttons
  - **OFF** button
  - **STANDALONE** button



Status bar
Displays pairing status, wheel-lock status, patient position, and battery life.

LCD touch display
Displays current table positioning, as well as system status messages.

Da Vinci button
Pressing initiates pairing and to turn on/turn off Integrated Table Motion during procedures.

Positioning
Adjusts table height and position. Once any cannulas are docked, turn on Integrated Table Motion to enable these buttons.

Off
Powers the table off, or commands stop of any table movement.

Standalone
Allows standalone use of the table when no cannulas are docked.

Technology Overview

Standalone Mode Overview

Learning objectives

- Define Standalone mode
- Demonstrate how to enter Standalone mode

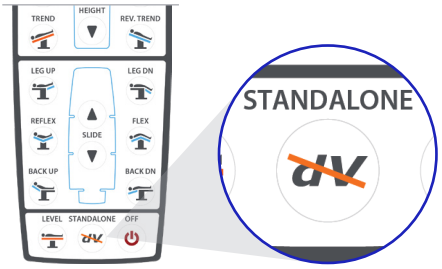
Facilitator note Ensure that the patient cart is not docked to the Skills Model during this activity

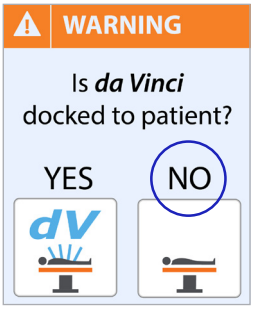
Enter Standalone Mode Activity

In an undocked and unpaired state, the TS7000dV table can be positioned in Standalone mode if the user first confirms the undocked status.

OR staff and anesthesiologist

1. Press and release the **STANDALONE** button on the TS7000dV remote control.
2. Read the message on the remote control's display: **Is da Vinci docked to patient?**
3. Visually confirm that the da Vinci Xi system is not docked to the patient, and select **NO** on the display.
4. Observe that the positioning buttons become lit and are available to move the table.





Enter Standalone Mode Discussion

- If a compatible system is not detected, only the **STANDALONE** button will be lit on the remote control.
- When attempting to enter Standalone mode, if the user indicates that the system is docked to the patient (i.e. by selecting **YES** on the remote control), no positioning buttons will be available or lit, and a message will appear accordingly on the remote control's display.
- Note that once Standalone mode times out, the user must re-confirm the undocked status in order to make standalone table adjustments.

Demonstrate Table Adjustments in Standalone Mode

Learning objectives

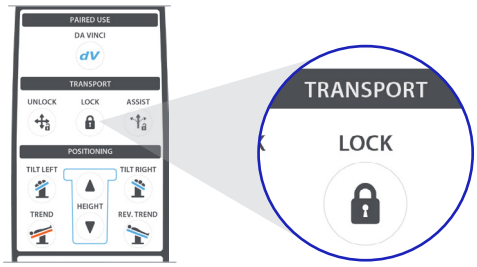
- Demonstrate locking the table feet
- Demonstrate table height adjustment
- Define iso-center movement
- Demonstrate Trend (with iso-center movement), tilt, and slide adjustments

Facilitator note Ensure that the table feet are not locked prior to the following demonstration

Lock Table Activity

OR staff and anesthesiologist

- Press and hold the **LOCK** button on the TS7000dV remote control to lock the table's feet
- A locked icon will appear on the remote control LCD status bar

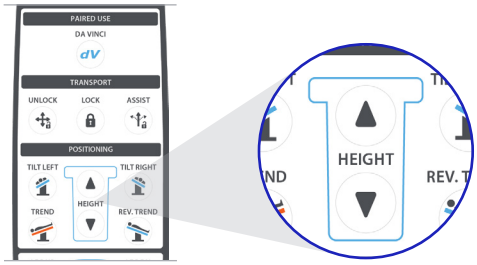


Warning: Always lock the table wheels prior to docking and do not unlock the brakes until the table is undocked. An unlocked operating table could unintentionally move during the procedure causing procedure conversion or harm to the patient or user.

Lower the Table Activity

OR staff and anesthesiologist

1. Enter the table's Standalone mode.
2. Press and hold the **HEIGHT ▼** button on the remote control to begin lowering the table. Note that the displayed height is rounded to the nearest whole number.
3. Demonstrate that once the table is lowered to a height of 24.8" (63 cm), its speed decreases, but the table can continue to lower until 22.8" (58 cm).
4. Set the table height to 26" (66 cm) to continue with this in-service.



Technology Overview

Iso-Center Movement Discussion

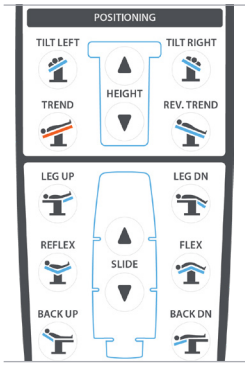


- Iso-center movement is a feature unique to the TS7000dV table
- Trendelenburg adjustments utilize a combination of Trend and slide functions, creating a virtual pivot point about which the table moves; this concept is analogous to a da Vinci cannula's motion about its remote center
- To take full advantage of iso-center motion, the patient should be positioned such that the ports are in-line with the table column, thereby:
  - Minimizing port travel in the cephalad-caudal direction during Trendelenburg adjustments
  - Minimizing port-site tension during table movement
  - Maximizing the range of available table motion while docked (by preventing the da Vinci Xi from reaching range-of-motion limits during ITM)

Facilitator note Emphasize the benefit of iso-center movement during trendelenburg adjustments.

Demonstrate Standalone Adjustment — Trend Activity

OR staff and anesthesiologist



For the demonstration, place a finger on a cannula to hold it upright during table movement.

1. Enter the table's Standalone mode
2. Press and hold the **TREND** positioning button on the remote control, and demonstrate the following:
   a. The table moves with a combination of Trend and slide motions, with the corresponding angle and position changes indicated on the remote control's LCD display.
   b. Because of iso-center movement, port position does not change appreciably, even during a large Trendelenburg adjustment (0°-25°).
   c. The table's Trend movement will pause when passing through 0° (when level).

Demonstrate Standalone Adjustments — Slide and Tilt Activity

OR staff and anesthesiologist

1. Enter table's Standalone mode
2. Demonstrate adjustment of table slide by using the **SLIDE** positioning buttons on the remote control
3. Demonstrate adjustment of table tilt by using the **TILT LEFT** and **TILT RIGHT** positioning buttons on the remote control
   - The table's tilt movement will pause at 15° and when passing through 0° (when level)

Note: There is no iso-center motion for tilt, but the da Vinci Xi system slide joints and boom can handle large changes in tilt.

2 Prepare for Intraoperative Use

Optimizing Setup — Preparing the Table

Learning objectives

- Define pairing, demonstrate wireless pairing process, and describe pairing best practices.
- Demonstrate table preparation for patient transfer, including patient orientation selection and table leveling.

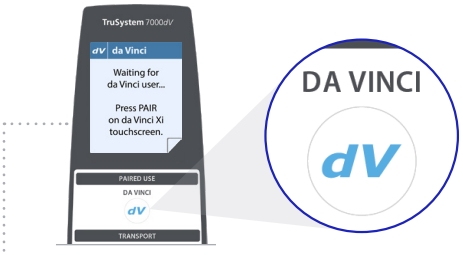
Pairing Overview Discussion

- Pairing is a two-step process that establishes a unique communication link between one da Vinci Xi system and one TS7000dV table.
- Pairing can be established wirelessly or through a wired connection.
- Pairing is required to turn on ITM.
  - Being in a paired state does not mean that ITM is turned on.
- Pair the da Vinci Xi to the TS7000dV table as soon as the system is powered on, so that it can monitor if cannulas are docked, and allow for safe movement of the table.
  - In an undocked, paired state, all positioning buttons will become active.

Wireless Pairing Activity

Surgeon, OR staff, and anesthesiologist

1. Power on the da Vinci Xi system.
2. Demonstrate that the TS7000dV remote control illuminates both the **DA VINCI (dV)** and **STANDALONE** buttons:
   a. The **DA VINCI (dV)** button illuminates as the TS7000dV table detects a compatible da Vinci Xi system.
   b. The **STANDALONE** button remains lit and available for standalone use after confirmation of undocked status (and until the system and table are paired).
3. Press the **DA VINCI (dV)** button on the TS7000dV remote control.
4. Within 10 seconds, tap **PAIR** on the vision cart touchscreen.
   a. After 10 seconds, the option to pair will time out, and the user will need to press the **DA VINCI (dV)** button again to establish pairing.



Prepare for Intraoperative Use

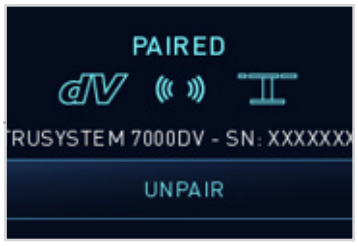
Wireless Pairing Activity Continued

5. Confirm that the system and table are paired:
   a. The TS7000dV remote control status bar displays a wireless connection icon
   b. The vision cart touchscreen's table tab shows the word **PAIRED** and three blue icons, representing the system, the paired wireless connection, and the table
   c. The table icons in the endoscope status area of the vision cart touchscreen and surgeon console 3D viewer are illuminated
      - If wireless pairing is not available or permitted at a given hospital, refer to this guide's troubleshooting section for steps on pairing using a wired connection

Note: At any time, to unpair, tap **UNPAIR** in the table tab.









Caution: Always verify the connection status (pairing) between the da Vinci Xi system and the TS7000dV table before performing Integrated Table Motion to prevent patient injury.

Prepare for Intraoperative Use

Facilitator note Ensure the TS7000dV remote control is connected to the receptacle at the head-end of the table column.

Lock Table, Select Patient Orientation, and Level the Table Activity

OR staff and anesthesiologist

1. Refer to the TS7000dV remote control's status bar to ensure that the table is locked. If unlocked, press and hold the **LOCK** button to lower and lock the table feet.
2. Select patient orientation as follows:
   a. Identify the head and foot sections of the table.
   b. Typically, the patient will be positioned with their head over the head section of the table. However, if the patient will be positioned in the opposite direction (i.e. with their head over the table's foot section), select this reverse orientation on the table's column keypad.
   c. Point out the patient orientation icon on the remote control's status bar. If a reverse orientation has been selected on the table column keypad, observe the consequent icon change on the status bar, reflecting the patient's inverted direction as seen from the anesthesiologist's point of view.
3. Level the table to facilitate patient transfer:
   a. Press and hold the **LEVEL** button on the remote control until the table is leveled and centered.
      - The table will level itself with respect to the floor, and will slide the tabletop so the back section is centered over the column.
   b. Observe the remote control display, and note audible feedback upon completion.

Note: When leveling the table, the tilt and Trend angles will return to 0°, while the height remains as set by the user.





Optimizing Setup — Preparing the Patient

Learning objectives

- Describe recommendations for positioning and securing the patient
- Describe recommendations for port placement

Facilitator note Ensure that the da Vinci Skills Model is positioned and secured per instructions at the beginning of this guidetable column.

Position and Secure the Patient Discussion

OR staff and anesthesiologist

1. If possible, position the patient such that the ports are in-line with the table column to take full advantage of iso-center movement.
2. Properly secure the patient to the table to avoid unintended movement during the procedure.
3. Do not wrap tape or straps underneath the tabletop as they may interfere with the column when the table slides.
4. If available, attach low-profile stirrups to increase clearance between the patient and the table.
5. Test all planned table positions to ensure that the patient does not slide on the table.



Warning: Always follow hospital protocol to ensure that the patient is secured such that they can reach the intended patient positions without shifting to prevent patient tissue damage.

Facilitator note Ensure proper port placement through ITM silicone grommets per instructions at the beginning of this guide.

Place Ports Discussion

Surgeon and OR staff

1. Place ports for the procedure, following guidelines in the da Vinci Xi System User Manual.
2. Ensure that the cannulas are adequately retained to prevent unwanted motion during ITM.
   - Only compatible multi-port accessories can be used. ITM is not compatible with flared cannulas or with da Vinci Single-Site curved cannulas and instruments.
3. Set table height at approximately 25 - 27″ (65 - 68 cm) to ensure adequate patient clearance for da Vinci Xi arms during docking and targeting.

Caution: Integrated Table Motion should not be used in procedures where cannulas are not fully retained by a body wall (for example, access through oral cavity), cannulas are placed through the chest wall (for example, cardio-thoracic procedures), or if two or more cannulas are collocated in the same abdominal access port, to prevent patient harm.

Optimizing Setup — Preparing the Patient Cart

Learning objectives

- Explain and demonstrate Guided Setup process, emphasizing best practices to maximize range-of-motion for ITM.
- Demonstrate how the patient cart arms move when brakes are disengaged, and review arm range-of-motion limits as relevant to ITM.

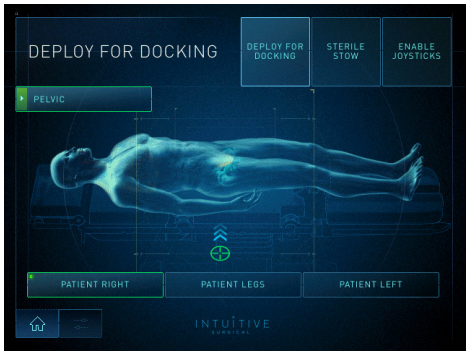
Facilitator note Emphasize that deploying for docking via Guided Setup will maximize the range of motion for the boom during ITM.

Deploy and Drive Patient Cart Activity

Surgeon and OR staff

Follow messages on the vision cart touchscreen's Guided Setup tab:

1. On the patient cart helm touchpad, select anatomy and patient cart approach
   a. Do not select **ORAL**, **NECK**, **THORACIC-TRANSVERSE**, or **THORACIC-SAGITTAL** anatomy as this will disable ITM
2. Press and hold **DEPLOY FOR DOCKING** until the boom finishes repositioning
3. Drive the patient cart to the table, and position the laser crosshairs within 5 cm of the endoscope port



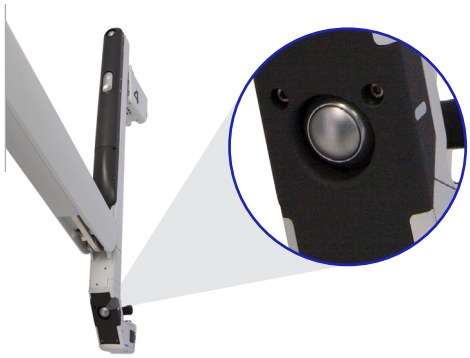
Note: For best results with ITM, use Guided Setup to deploy for docking; doing so will allow optimal range of motion for the boom and patient cart arms as they follow port movement during ITM (referred to as port following)

Warning: Do not perform Integrated Table Motion when using inadequately retained cannulas (for example, in situations that reduce the pullout force such as procedures through the chest wall, procedures without a body wall, or patients with overly compliant/weak/thin/fatty body walls). If necessary, undock the affected instrument arm(s) prior to performing Integrated Table Motion to prevent loss of insufflation, patient harm, and tissue injury.

Arm Manipulation and Range-of-Motion Limits Activity

Surgeon, OR staff, and anesthesiologist

1. Using the port clutch button, manipulate a patient cart arm to experience the force required to move the arm when its brakes are disengaged (as will be the case during ITM)
2. Move an arm along its vertical and slide joints, and identify the red markings that indicate range-of-motion limits



Note: During ITM, if the system or table reaches a limit, only further table movement in the direction of the limit will be impacted

Prepare for Intraoperative Use

Facilitator notes

- Confirm paired status, emphasizing that the da Vinci Xi monitors if cannulas are docked to allow safe table movement
- Facilitate demonstration of incomplete targeting by increasing table height or by setting a low boom height limit
- Emphasize the use of ITM to adjust table height when targeting fails

Docking and Targeting Activity

Surgeon and OR staff

1. Dock arm 2 as the initial endoscope arm.
   a. Once an arm is docked, only the **DA VINCI (dV)** button will be available (lit) on the remote control until ITM is turned on.
   b. When 1 or 2 cannulas are docked, only table height can be adjusted using ITM.
2. Discuss and perform targeting:
   a. Targeting centers the boom and patient cart arms over the surgical workspace, maximizing the range of table positions available to the surgeon when using ITM intraoperatively.
3. Demonstrate incomplete targeting, and show that ITM can be used to lower the table as follows:
   - Press the **DA VINCI (dV)** button on the remote control to turn on ITM, and use the illuminated height positioning buttons to adjust table height. Press the button again to turn off ITM and re-engage the patient cart arm brakes.
4. Reattempt targeting until targeting is complete.
5. Dock the remaining arms and adjust arm spacing following Guided Setup best practices.
6. Confirm that there is sufficient slack in cables and tubing to accommodate table movement and to ensure instrument arm movement is not restricted during Integrated Table Motion.

Note: After docking, the arrangement of the system, table, and Skills Model should match the setup displayed here:



3 Intraoperative Use

ITM — Without Instruments

Learning objectives

- Demonstrate preparation for ITM
- Describe audible and visual indicators of ITM
- Demonstrate turning ITM on and off (without installed instruments), emphasizing audible and visual indicators of ITM
- Describe safety considerations before, during, and after ITM

Facilitator note Due to the close placement and nature of the silicone ITM grommets on the Skills Model, visualize the cannula tip furthest from the endoscope. To facilitate endoscopic visualization of other cannula tips, consider port hopping with the endoscope.

Prepare for ITM Activity

Surgeon, OR staff, and anesthesiologist

1. Confirm that the system and table are paired.
2. Use the endoscope to visually confirm cannula tip clearance from internal anatomy to avoid inadvertent contact with tissue during ITM.
3. Check patient clearance from instrument arms and external obstacles (e.g. stirrups).
4. To prevent patient harm or tissue injury, check cannulas for tension, and release any tension using the port clutch buttons.
5. Confirm that all patient cart LEDs are solid blue. ITM cannot be turned on if the system is in a faulted state or if any of the docked arms are clutched.



Facilitator note Emphasize audible and visual indicators of ITM to facilitate their observation during the activity that follows.

Audible and Visual Indicators of ITM Discussion

During ITM, the following audible and visual indicators can be observed:

- Blue blinking LEDs on the patient cart arms
- Continuous chimes from the da Vinci Xi system
- Blue ITM banner on the vision cart and surgeon console, displaying a message: **READY FOR TABLE MOTION** or **TABLE MOTION IN PROGRESS**

The table's position can be monitored as follows:

- **Surgeon and OR staff:** Observe degrees of Trend and tilt in the endoscope status area on the vision cart touchscreen and surgeon console
  - Degrees of Trend and tilt are updated as the table moves during ITM
- **Anesthesiologist:** Observe indicators of table position (i.e. degrees of Trend and tilt, slide displacement, height, etc.) on the TS7000dV remote control display
  - Values are updated as the table moves

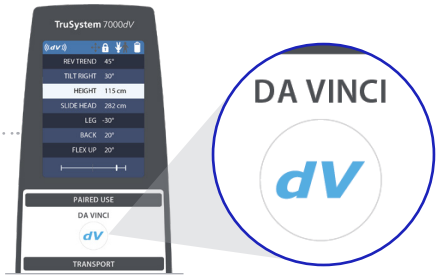
Intraoperative Use

Facilitator notes

- To facilitate observation of boom movement during port following, consider positioning at least one patient cart arm such that it reaches its slide joint motion limit during ITM
- In the workflow below, assume that tasks may apply to all roles unless indicated as role-specific actions

Turn on ITM — Without Instruments Activity

Demonstrate read-back technique to communicate all table movements with the surgeon, anesthesiologist, and patient-side assistant.



1. Turn on ITM:
   a. **Anesthesiologist:** Press the **DA VINCI (dV)** button once to turn on ITM
   b. **OR staff:** Observe audible clicks that indicate the da Vinci Xi system is releasing its arm brakes, allowing the arms to follow the ports when the table is repositioned
   c. **Anesthesiologist:** Observe available positioning buttons illuminate on the remote control
2. Demonstrate table tracking:
   a. **Anesthesiologist:** Adjust table position using the **TREND** positioning button
   b. **OR staff:** Observe patient cart arm movements that track with the table, maintaining the endoscope orientation and the angle of the ports relative to the body wall, thereby minimizing port-site strain
3. Demonstrate port following:
   a. **Anesthesiologist:** Adjust table position using the **SLIDE** positioning button
   b. **OR staff:** Observe that if an arm reaches a range-of-motion limit, the boom will move as needed for port following
4. As the anesthesiologist continues to make table adjustments, observe audible and visual indicators of ITM and monitor table movement as follows:
   a. **OR staff:** Blue blinking LEDs on patient cart arms, continuous system chimes, and blue ITM banner message (**READY FOR TABLE MOTION** or **TABLE MOTION IN PROGRESS**)



   b. **Surgeon and OR staff:** Observe degrees of Trend and tilt displayed in the endoscope status area, updated as the table moves

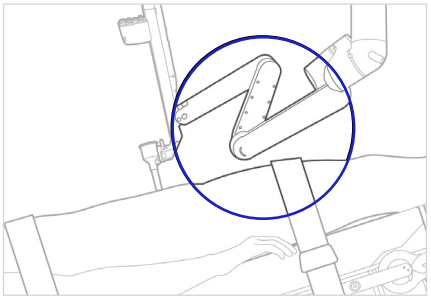


   c. **Anesthesiologist:** Monitor changes in table position (i.e. degrees of Trend and tilt, slide, height, etc.) indicated on the remote control display

Intraoperative Use

Turn off ITM — Without Instruments Activity

1. Turn off ITM:
   a. **Anesthesiologist:** After table adjustments have been made using ITM, press the **DA VINCI (dV)** button once to turn ITM off.
   b. **OR staff:** Observe the audible clicks of the patient cart arm brakes re-engaging, as well as a voice prompt: **Check patient clearance**. Note that the blue ITM banner disappears.
2. Demonstrate ITM timing out:
   a. **Anesthesiologist:** Turn on ITM again, and press any positioning buttons to adjust the table.
   b. **OR staff:** As ITM times out after a brief period, observe the re-engagement of the patient cart arm brakes, as well as the voice prompt: **Check patient clearance.**



Check patient clearance

Safety Checks Before, During, and After ITM Activity

OR staff

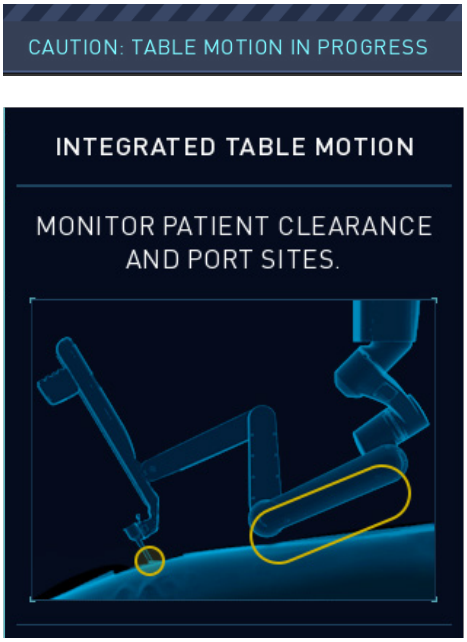
1. Check patient clearance:

Warning: Check for clearance between the patient and the instrument arms before, during, and after Integrated Table Motion. Prolonged pressure on the patient could result in serious injury.

   a. The da Vinci Xi system monitors the location of the patient cart arms, cannulas, and the table, but it cannot monitor the patient's position or stirrups.
   b. The **check patient clearance** prompt is a reminder for patient-side staff to manually check for patient clearance after every occurrence of table movement with ITM.
2. Confirm adequate space between the patient, ancillary equipment, and/or instrument arms.

Warning: Watch for potential contact between the patient and the da Vinci Xi system or other equipment (such as instrument arms) during Integrated Table Motion to prevent patient injury.

3. If inadequate patient clearance is observed after ITM, adjust the patient clearance joints or flex joints as needed.
4. To prevent patient harm or tissue injury, check port sites to ensure proper placement of cannula remote centers within the patient's body wall, and release tension if required.
   a. Release tension with caution as this can move the instrument tips.



Message on vision cart monitor during ITM

Intraoperative Use

ITM — With Instruments

Learning objectives

- Demonstrate attempt to turn on ITM when three instruments are installed
- Demonstrate turning on ITM when two instruments are installed and actively controlled

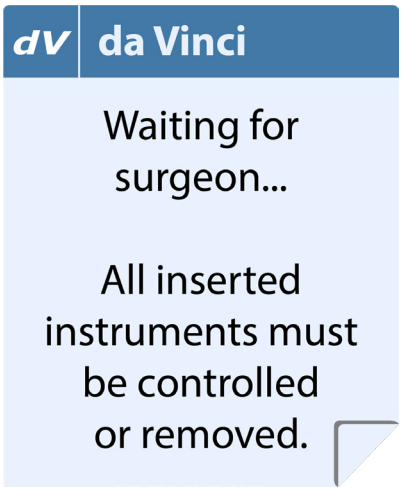
Facilitator notes

- To prepare for this activity, ask the surgeon and sterile OR staff participants to install three instruments on the patient cart arms and to coordinate insertion of all instruments into the Skills Model under direct visualization. Ask the surgeon participant to remain seated at the surgeon console.
- Emphasize the requirement that all instruments must be actively controlled and visualized in order to turn on ITM.

Attempt to Turn on ITM — With Three Instruments Activity

1. Surgeon: At the surgeon console, match grips with two active instruments. Keep instrument tips in view, and position them to avoid inadvertent contact with tissue.
2. Anesthesiologist: Attempt to turn on ITM by pressing the **DA VINCI (dV)** button on the TS7000dV remote control.
3. Surgeon, OR staff, and anesthesiologist observe:
   - Da Vinci Xi audible feedback.
   - Vision cart monitor yellow banner message: **NOT READY FOR TABLE MOTION. INSERTED INSTRUMENTS MUST BE CONTROLLED FROM SURGEON CONSOLE OR REMOVED.**
   - Remote control message: **All inserted instruments must be controlled or removed.**
4. Surgeon: Using read-back technique, ask the patient-side assistant to remove one of the three instruments.
   - If a dual console is available, ask another surgeon to take control of the third instrument from their surgeon console.





Intraoperative Use

Facilitator notes

- Role-play as needed to facilitate participants' practice using ITM in this activity
- Emphasize that Guided Tool Change is deactivated after ITM

Turn on ITM — With Two Instruments Activity

Demonstrate using ITM to position the table with Trend, tilt, slide, and height adjustments, while practicing read-back technique:

- Anesthesiologist:
  - Press the **DA VINCI (dV)** button once to turn on ITM
  - Adjust table position as requested by the surgeon
  - Press the **DA VINCI (dV)** button once again to turn off ITM
- Surgeon:
  - Position the instruments to avoid inadvertent tissue contact
  - Maintain control and visualization of instrument tips while the table is moving
  - Avoid precise surgical tasks during ITM
  - Observe degrees of Trend and tilt displayed in the endoscope status area of the surgeon console 3D viewer
  - Pause ITM if necessary (see next section)

Caution: Continuously control and visualize instrument tips while the table is moving to prevent patient tissue injury.

- OR staff:
  - Continuously monitor clearance between the patient and the da Vinci Xi arms, as well as cable slack. Adjust as needed before, during, and after ITM.
  - Following ITM, demonstrate that Guided Tool Change is deactivated. Re-install the third instrument from the previous activity under direct visualization.

Intraoperative Use

Intraoperative Adjustments

Learning objective

Describe and demonstrate conditions requiring adjustment of port position.

Adjust Port Position Discussion

The following scenarios will stop ITM and prompt for the adjustment of port position:

- A range-of-motion limit is met, causing tension on the port
- An arm is interfering with another arm or with the patient
- A cannula is not well-retained or is moving differently from other cannulas

Warning: During Integrated Table Motion, it is necessary to recognize a range-of-motion limit and to resolve it to prevent patient harm or tissue injury due to high body wall forces.

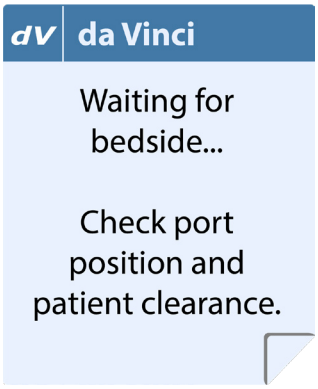
Facilitator notes

- Remove all instruments prior to this activity.
- Perform Steps 1 and 2 on the cannula furthest from the endoscope port to facilitate the visualization step.
- To demonstrate unexpected cannula movement and generate the prompt to adjust port position on a given arm, move a cannula slightly out of the Skills Model while the table and other arms move normally during ITM. (Do not pull the cannula all the way out).
- Demonstrate Steps 1 and 2, but speak to resolving the scenario in Step 3.

Adjust Port Position Activity

Surgeon, OR staff, and anesthesiologist

1. Demonstrate unexpected cannula movement:
   a. Anesthesiologist: Turn on ITM, and adjust table position using the **TREND** button
   b. Demonstrate that if a cannula moves in an unexpected direction (relative to other cannulas), the system will stop ITM and re-engage the patient cart arm brakes
   c. Observe error messages on the vision cart, surgeon console, and TS7000dV remote control display



TS7000dV remote control message

Intraoperative Use

Adjust Port Position Activity Continued

2. Adjust port position:
   a. Gain direct visualization of the cannula at which the unexpected movement occurred
   b. Use one hand to support the cannula at the body wall
   c. Use the other hand to press the port clutch button and realign the remote center in the body wall
3. Address the scenario in which a cannula has completely slipped out of the body wall:
   a. Remove any installed instrument(s) from the affected patient cart arm(s)
   b. Undock the cannula from the affected patient cart arm
   c. Re-insert the cannula in the body wall using standard surgical procedures
   d. Realign the cannula remote center within the body wall
   e. Re-dock the patient cart arm to the cannula, and release any port-site tension
   f. If applicable, re-install instrument(s) on the patient cart arm(s), and insert under visualization



Vision cart touchscreen prompt

Intraoperative Use

Pausing and Canceling ITM

Learning objectives

- Describe and demonstrate pausing ITM
- Describe and demonstrate canceling ITM

Pause ITM Discussion

When ITM is paused, table movement stops, but the patient cart arm brakes remain disengaged.

Facilitator notes

- Facilitate tasks (i.e. turning ITM on) as needed to demonstrate each role's ability to pause ITM
- Facilitate demonstration of the surgeon's ability to pause ITM by installing and inserting instruments before the following activity

Pause ITM Activity

OR Staff

Note: Due to radio frequency (RF) interference, using RF electrocautery during table movement can pause ITM

Anesthesiologist

- Pause ITM from the remote control by releasing depressed positioning button
- Observe that when ITM is paused by the surgeon, the TS7000dV remote control displays a message: **Waiting for da Vinci user.... Table motion is paused.**

Surgeon

- Pause ITM from the surgeon console by performing an action that interrupts active instrument control, such as:
  - Pressing the endoscope control pedal
  - Pressing the clutch pedal
  - Using the finger clutch
  - Removing head from 3D viewer
- Observe that when ITM is paused, the ITM banner turns yellow and displays a message: **TABLE MOTION PAUSED**





Intraoperative Use

Cancel ITM Discussion

When ITM is canceled, table movement stops, and the patient cart arm brakes re-engage (i.e. the arm brakes are applied).

- When instrument arm brakes are not applied, the instrument arms can drift during table motion, leading to interferences, breaches in sterility and change of arm positions.

Facilitator note Facilitate tasks (i.e. turning ITM on, recovering faults) as needed to demonstrate each role's ability to cancel ITM.

Cancel ITM Activity

Surgeon

- Cancel ITM from the surgeon console's right-side pod by pressing the Emergency STOP button, triggering a recoverable system fault



Anesthesiologist

- Cancel ITM by pressing the **DA VINCI (dV)** button on the TS7000dV remote control

OR Staff

- Cancel ITM by pressing a button on a patient cart arm, such as:
  - Pressing the port clutch button
  - Pressing the patient clearance button
  - Pressing the instrument clutch button
- Cancel ITM by installing an instrument on a patient cart arm
- Cancel ITM from the patient cart helm or from the surgeon console's right-side pod by pressing the Emergency STOP button, triggering a recoverable system fault

Caution: If you experience unintuitive motion during table motion, press the endoscope control pedal or stop Integrated Table Motion by pressing e-stop to prevent user confusion or tissue injury due to unintuitive motion.

4 Troubleshooting

Learning objective

Recognize and recover from common troubleshooting scenarios such as loss of pairing, pairing via a wired connection, and gaining immediate patient access.

Facilitator notes

To simulate a pairing lost scenario with a recoverable fault, while in a paired, docked, state:

1. Power down the table using the **OFF** button on the remote control
2. Observe recoverable fault (do not recover)
3. Return to the fault tab

Pairing Lost Activity

Surgeon, OR staff, and anesthesiologist

If pairing is lost while the arms are docked to the patient, a recoverable system fault is triggered:

1. Observe the message on the vision cart touchscreen indicating that the table is not paired with the system and that Integrated Table Motion is not allowed.
2. Recover the fault.
3. Press the **DA VINCI (dV)** button on the remote control to initiate a new pairing request. Then, tap **PAIR** on the vision cart touchscreen's Table tab to pair the system and table.
4. If pairing still fails, attempt to perform the following:
   a. Check the Table tab on the vision cart touchscreen for additional information.
   b. If attempting to pair wirelessly, ensure that all IR sensors are unblocked.
   c. Move or power off any nearby tables.
   d. Pair the system and table using a wired connection.
5. If pairing is still not possible, undock all arms, and only move the table in Standalone mode.

Facilitator notes

- Ensure availability of the wired connection cable for the following activity
- Terminate the wireless pairing connection from the previous activity to ensure an unpaired state

Establish Wired Connection Activity — Optional

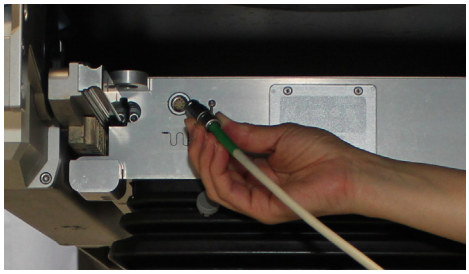
Surgeon, OR staff, and anesthesiologist

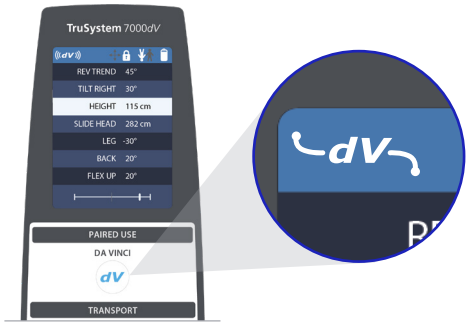
Use the optional connection cable to link the table and patient cart directly, bypassing IR/RF communication:

1. Connect one end of the cable to the patient cart jack.
   a. The jack is located at the back of the patient cart next to the blue fiber optic cable connection.
2. Connect the other end of the cable to either of the two table ports.
   a. The table ports are located at the top of the table column, just under the tabletop.
3. Confirm the system-to-table wired pairing using the remote control:
   a. A wired connection icon will appear on the remote control's status bar.
   b. The Table tab will display the word **PAIRED**, along with three blue icons representing the da Vinci Xi system, the wired connection, and the table.
4. Manage connection cable; do not run over the cable with the patient cart.

Note: At any time, to unpair, unplug the cable. Once unpaired, the table reverts to standalone use.







Troubleshooting

Facilitator note Emphasize best practice in a scenario necessitating immediate patient access or conversion to open surgery.

Gain Immediate Patient Access Discussion

If immediate access to the patient is required, perform the following steps to remove the system from the patient before moving the table:

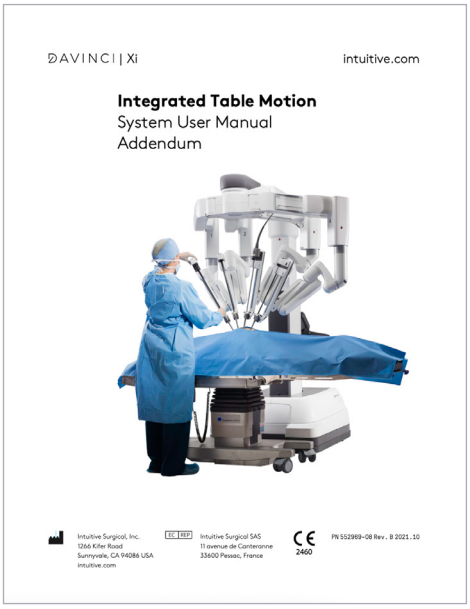
1. Check that instruments are not grasping tissue, and remove the instruments from the patient. If needed, use the instrument release kit (IRK) to manually release grips before removing the instruments.
2. Remove the endoscope from the patient.
3. Disconnect cannulas from the instrument arms.
4. Move the instrument arms away from the patient.
5. If necessary, move the patient cart away from the patient.
6. Once all arms are undocked, the TS7000dV remote control positioning buttons should illuminate for use.
    a. If positioning buttons are not lit after undocking all arms, press the **STANDALONE** button on the remote control, and follow prompts to move the table.
7. If the remote control is inoperable, confirm that all arms are undocked and moved away from the patient. Then, use the table's column keypad controls to enable table movement, as per instructions in the ITM User Manual Addendum.

Resources

Troubleshooting Appendix:

Refer to the Integrated Table Motion System User Manual Addendum for additional troubleshooting guidance on the following scenarios:

- Unable to pair the system and table
- Table range-of-motion limit reached
- Paired to the wrong system, wireless signal blocked, or not paired with da Vinci



Review Additional Resources:

- Da Vinci Xi System User Manual
- Integrated Table Motion User Manual Addendum
- Integrated Table Motion OR Reference Guide
- Baxter TS7000dV Operating Table Instruction Manual

Support:

- 24/7 Intuitive Surgical Technical Support:
  - US: 1.800.876.1310
  - Europe: +33.1.77.68.88.45
- Baxter Technical Support:
  - Email: HRC_Trumpf_tech_support@baxter.com
  - Phone:
    - US: 1.800.445.3720 (press 2; then, press 7)
    - EU: +49.3671.586-0
    - Canada: 1.800.267.2337 (calling from within Canada); 1.866.506.1355 (calling to Canada from US); 1.905.206.1355 (calling from outside US/Canada)

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